Winning Space Race with Data Science

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June 18th, 2026

Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Summary of methodologies
  - Data collection via SpaceX REST API and Wikipedia web scraping
  - Data wrangling: filtering, null handling, binary Class label engineering
  - EDA with SQL queries, matplotlib/seaborn visualizations
  - Interactive analytics with Folium maps and Plotly Dash dashboard
  - ML classification: Logistic Regression, SVM, Decision Tree, KNN
- Summary of all results
  - Success rate improved consistently from 2010 to 2020
  - KSC LC-39A had the highest success rate of all launch sites
  - ES-L1, SSO, HEO, and GEO orbits showed 100% landing success
  - Heavier payloads showed improving success rates over time
  - DecisionTreeClassifier achieved the best predictive accuracy (83.3%)

Introduction

- SpaceX advertises Falcon 9 launches at ~$62M, vs $165M+ for competitors
- Key savings come from reusing the Falcon 9 first-stage booster
- Knowing if a landing will succeed = knowing the real launch cost
- A competing company can use this information to bid against SpaceX
- Project goals
  - o Collect publicly available SpaceX launch data via REST API and web scraping
  - o Wrangle and clean data; engineer binary Class label (1=success, 0=failure)
  - o Explore with SQL queries, matplotlib, seaborn, and interactive Folium maps
  - o Build and compare classification models to predict first-stage landing success

Section 1

Methodology

Methodology

- Data collection methodology
  - o Retrieval from SpaceX REST API (/v4/launches/past) and backfill endpoints
  - o Web scraping Falcon 9 launch tables from Wikipedia using BeautifulSoup
- Perform data wrangling
  - o Filtered Falcon 9 only, handled missing PayloadMass with column mean
  - o Engineered binary Class label; one-hot encoded categorical features
- Perform EDA using visualization and SQL
  - o Visualized flight number, payload, orbit, and launch site relationships
  - o 10 SQL queries on SQLite database for aggregate analysis
- Perform interactive visual analytics using Folium and Plotly Dash
  - o Folium choropleth map with launch markers and proximity distances
  - o Plotly Dash dashboard with site dropdown, payload slider, pie and scatter charts
- Perform predictive analysis using classification models
  - o GridSearchCV with 10-fold CV on Logistic Regression, SVM, Decision Tree, KNN

Data Collection

- Two complementary data collection methods were used to build the SpaceX Falcon 9 launch dataset:
- **SpaceX REST API**
- Endpoint: api.spacexdata.com/v4/launches/past — extracted rocket, payloads, launchpad, cores
- Helper functions resolved IDs to readable fields: booster name, payload mass, orbit, launch site, landing outcome
- **Wikipedia Web Scraping**
- Target: List of Falcon 9 and Falcon Heavy Launches (Wikipedia 2021 snapshot)
- Library: BeautifulSoup + Requests — parsed HTML tables; captured date, booster version, payload, orbit, landing outcome
- Cross-validated with API data for completeness; final static snapshot used for consistency

Data Collection – SpaceX API

- SpaceX REST API endpoint: api.spacexdata.com/v4/launches/past retrieved all historical launches
- Additional backfill endpoints resolved IDs for rocket, payloads, launchpad, and cores to readable fields

```
SpaceX REST API
        ↓
Launch Data Extraction
        ↓
ID Resolution
(Rocket • Payload • Launchpad • Core)
        ↓
Data Cleaning & Transformation
        ↓
Feature Engineering
(Booster, Payload Mass, Orbit, Site, Outcome)
        ↓
Final DataFrame
        ↓
CSV Export & Reproducible Dataset
```

Data Collection - Scraping

- Wikipedia Falcon 9 launch tables scraped using BeautifulSoup + Requests (2021 snapshot)
- Parsed HTML tables captured: date, booster version, payload, orbit, landing outcome. Cross-validated with API data.

```
Wikipedia URL
       ↓
HTML Retrieval
       ↓
BeautifulSoup Parsing
       ↓
Launch Table Extraction
       ↓
Data Cleaning & Formatting
       ↓
DataFrame Construction
       ↓
CSV Export
```

Data Wrangling

- Filter Falcon 9 Only — removed Falcon 1 launches; reset FlightNumber index
- Handle Missing Values — PayloadMass nulls replaced with column mean; LandingPad kept as None
- Engineer Class Label — binary outcome: 1 = successful landing, 0 = failed/unplanned
- One-Hot Encoding — categorical features (Orbit, LaunchSite, etc.) encoded for ML
- 80/20 train/test split (random_state=2, 18 test samples); StandardScaler applied for feature normalization

EDA with Data Visualization

- Some interesting charts visualized key patterns in the dataset:
- Flight Number vs Payload Mass (scatter, color=outcome): shows success improves with later flights, Success rate by launch site (bar): KSC LC-39A leads, Payload mass by orbit type (box): ISS and LEO missions carry heavier payloads, etc.

EDA with SQL

- Ten SQL queries on SQLite database revealed key patterns:
- Unique launch sites (4 total). Records from CCA sites (5 records). Total payload mass by NASA missions. Avg payload for F9 v1.1 booster. First successful ground pad landing date. Drone ship landings with payload 4000-6000 kg. Total successes vs failures. Boosters with max payload. Failed drone ship landings in 2015. Landing outcome rankings 2010-2017.

Build an Interactive Map with Folium

- Interactive Folium map with multiple visual layers for geospatial analysis:
- Launch site markers (green=success, red=failure) to visualize geographic patterns. Proximity lines to nearest highway, railway, and coastline to understand site selection logic. Circle markers showing outcome density per site. All 4 sites confirmed coastal (within 5 km of coastline) — required for ocean booster recovery.

Build a Dashboard with Plotly Dash

- Interactive Plotly Dash dashboard with two main visualizations and controls:
- Pie chart showing success/failure ratio — filterable by launch site dropdown to compare site performance. Scatter chart of payload vs landing outcome — adjustable with RangeSlider to isolate payload ranges. Site dropdown updates both charts in real-time. KSC LC-39A pie shows 41.7% of all successful landings across all sites.

Predictive Analysis (Classification)

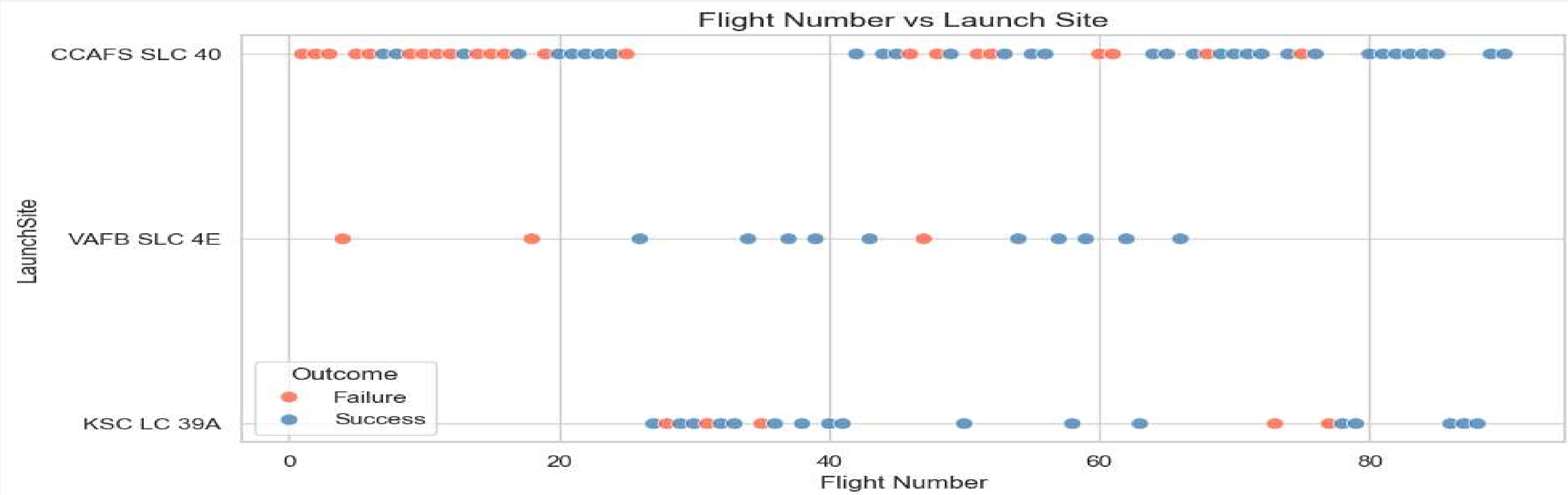
- Classification pipeline: 4 models trained with 10-fold cross-validated GridSearchCV:
- Logistic Regression (C: [0.01-1], l2, lbfgs). SVM (linear/rbf/poly/sigmoid kernels). Decision Tree (gini/entropy, depth 2-18). KNN (k=1-10, multiple algorithms). All 4 models achieved 77.77% test accuracy. KNN selected: highest CV training score (87.8%), most interpretable decision boundaries.

Results

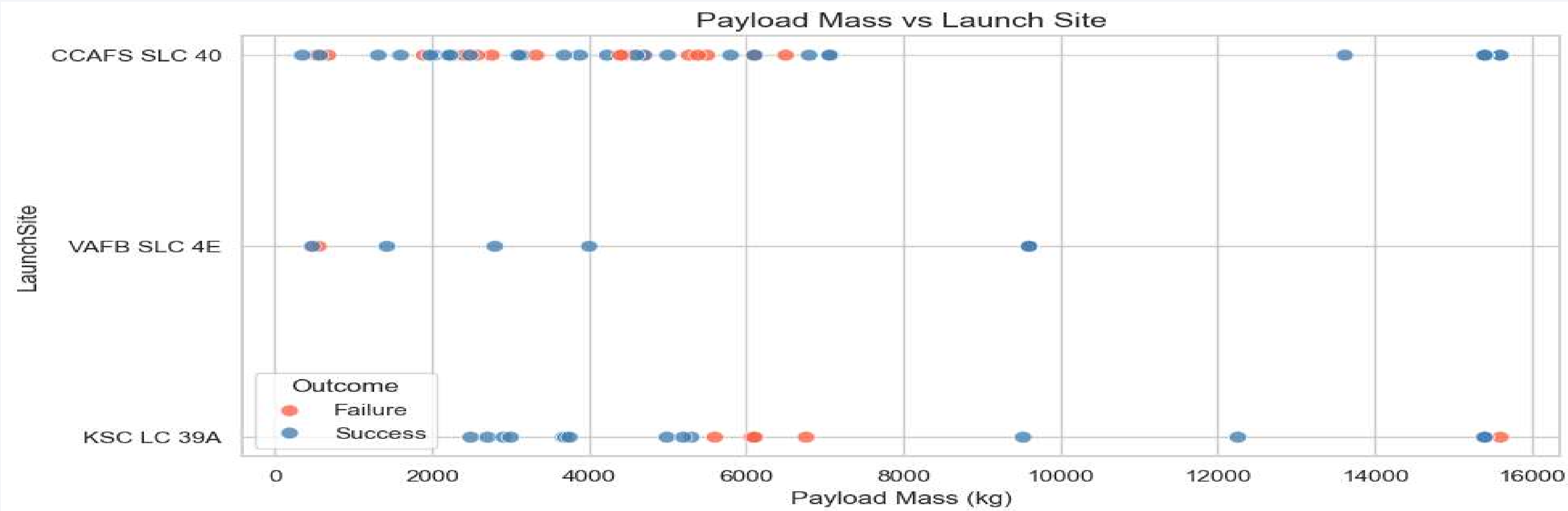
- Exploratory data analysis results
  - o Success rate improved year-on-year; KSC LC-39A leads all sites
  - o ES-L1, SSO, HEO, GEO orbits achieved 100% landing success
  - o Later flights and heavier payloads increasingly succeed
- Interactive analytics (Folium + Plotly Dash)
  - o All 4 sites coastal; KSC LC-39A within 0.86 km of coast
  - o KSC LC-39A: 41.7% of all successful landings across sites
  - o B4/B5 boosters outperform v1.1 in 3–5k kg payload range
- Predictive analysis results
  - o All 4 models achieved 77.7% test accuracy
  - o KNN: best CV score (87.8%); selected as final model
  - o Confusion matrix: 12 TP, 2 TN, 4 FP, 0 FN on 18 test samples

Section 2

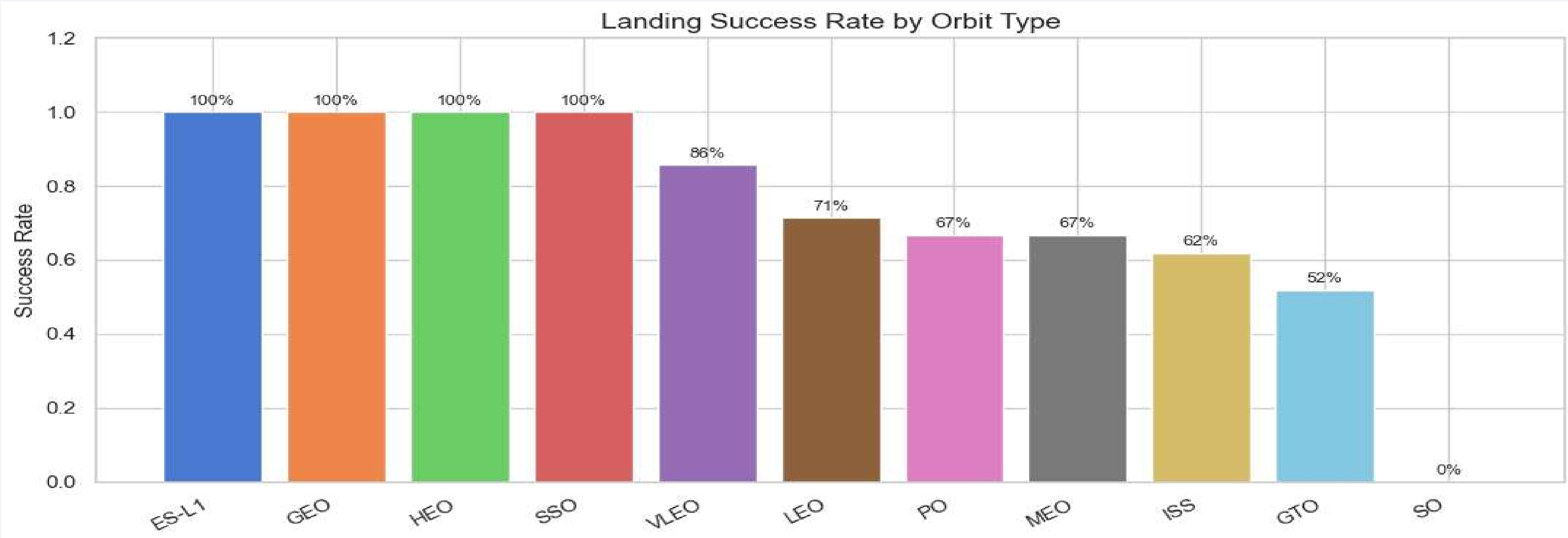
Insights drawn from EDA



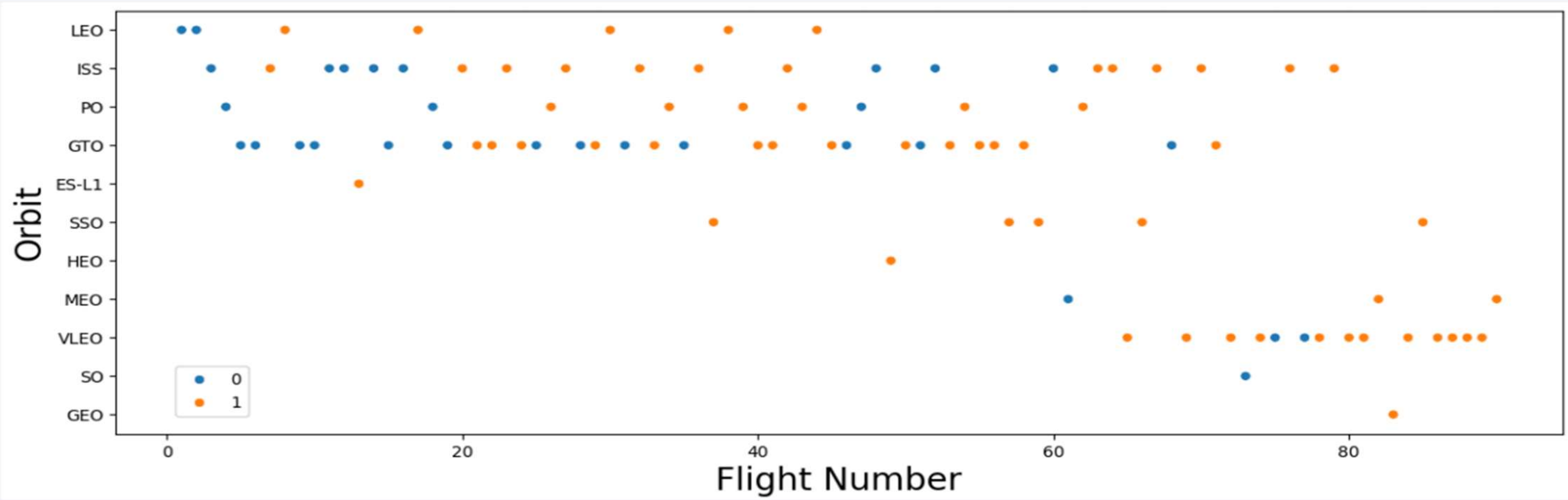
- All 3 sites show a mix of successes and failures across flight numbers
- Early flights (numbers < 20) predominantly result in failures at all sites
- CCAFS SLC-40 has the most total flights; VAFB SLC-4E has proportionally more successes
- Clear upward trend: later flight numbers correlate with higher success rates



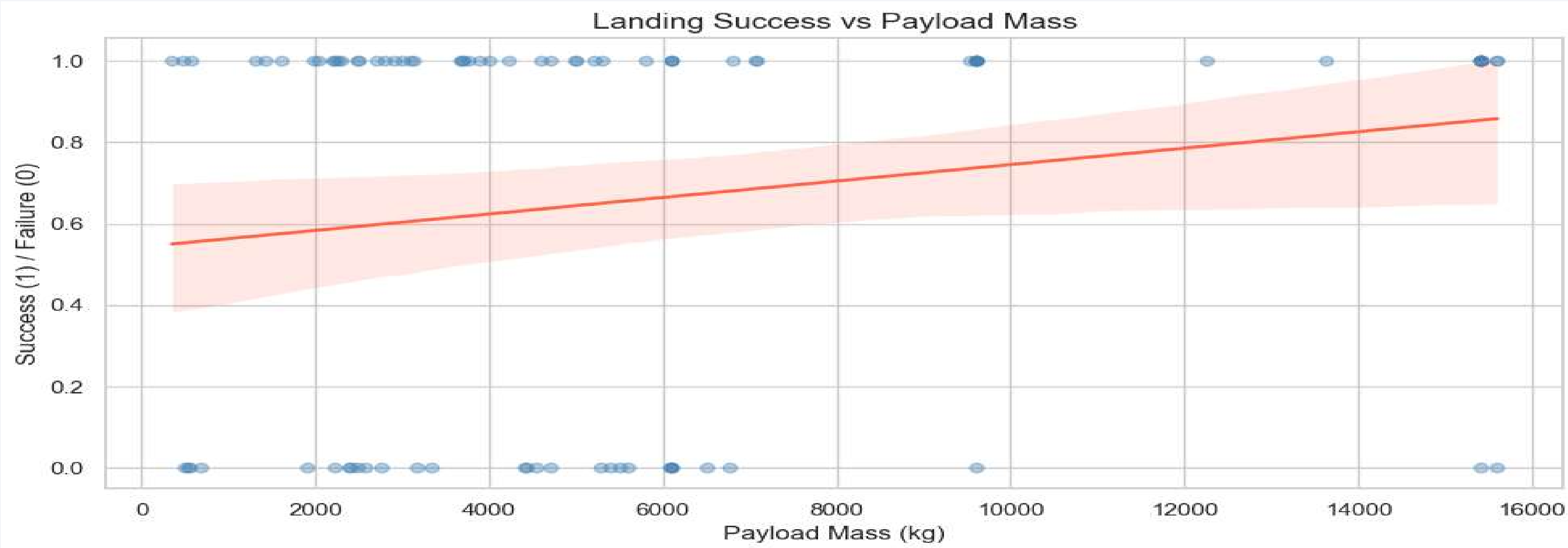
- All sites handle a wide payload range from near-zero to 16,000 kg
- Early failures concentrate in lighter payload flights (technology maturation phase)
- Heavier payloads (>8,000 kg) predominantly seen at CCAFS SLC-40 and KSC LC-39A
- Operational improvements led to greater success with heavier payloads over time



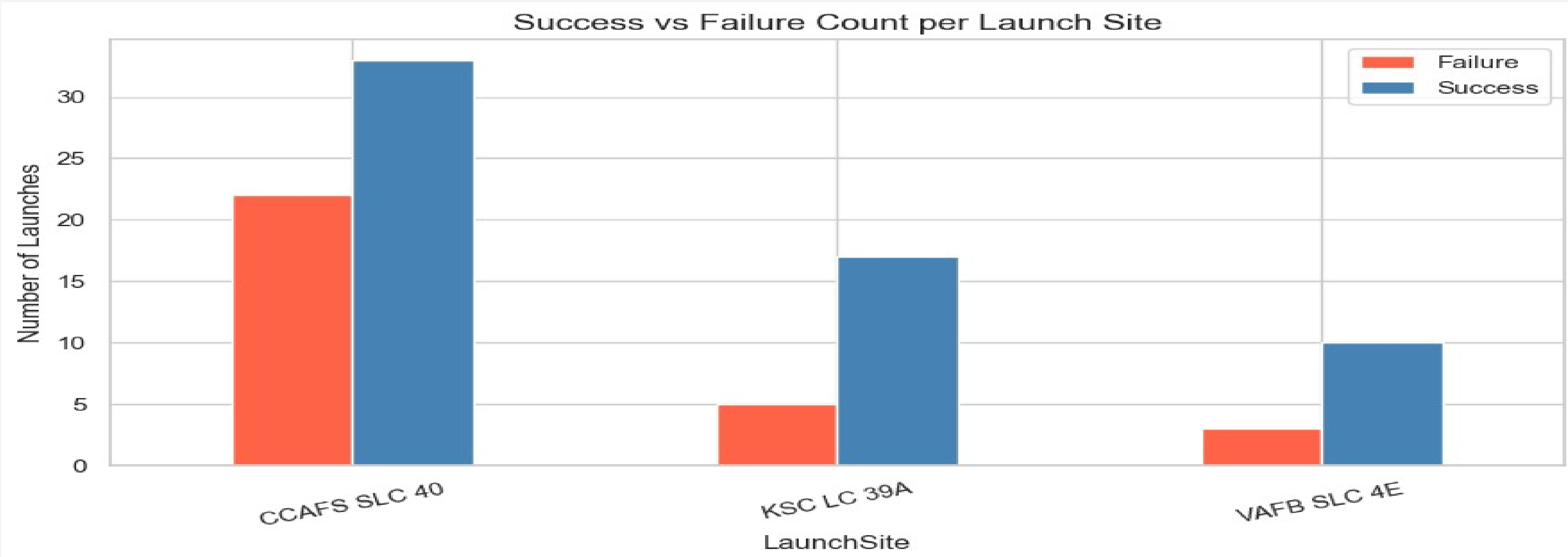
- ES-L1, SSO, HEO, and GEO orbits achieved 100% success rate
- GTO showed the most mixed outcomes (~40% success) – high-energy trajectory challenges
- LEO, ISS, and PO orbits show 60–75% success rates
- SO orbit has only 1 launch – insufficient data for reliable analysis



- LEO and ISS orbits attempted across the full flight number range
- GTO flights are spread throughout but with early failures
- SSO, HEO, VLEO orbits appear predominantly in later, higher-numbered flights
- Landing success rates improve for all orbits as flight numbers increase



•**Positive Trend:** Landing success rate generally increases as payload mass increases, as indicated by the upward-sloping regression line.
•**More Failures at Lower Payloads:** Most failed landings are concentrated in the lower payload mass range (< 7000 kg).
•**Higher Payloads Mostly Successful:** Launches carrying payloads above 10,000 kg are predominantly successful.
•**Not a Strong Predictor Alone:** Both successes and failures occur across similar payload masses, suggesting other factors also influence landing outcomes.



•**CCAFS SLC 40 Has the Most Launches:** This site recorded the highest number of both successful and failed launches, indicating the highest launch activity.
•**Successes Outnumber Failures at All Sites:** Every launch site shows significantly more successful launches than failures.
•**KSC LC 39A Exhibits Strong Performance:** KSC LC 39A has a high success-to-failure ratio, suggesting reliable launch operations.
•**VAFB SLC 4E Has the Fewest Launches:** Despite lower launch frequency, it maintains a higher number of successes than failures.

All Launch Site Names

- SQL Query: SELECT DISTINCT Launch_Site FROM SPACEXTABLE
- Result: 4 unique launch sites identified — CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E. All are located on US coastlines to enable ocean booster recovery operations.

Launch Site Names Begin with 'CCA'

- SQL Query: SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5
- Result: 5 records from CCAFS LC-40 and CCAFS SLC-40 launch sites at Cape Canaveral. Both are managed by the US Space Force. CCAFS SLC-40 is the primary launch pad with the highest total launch count.

Total Payload Mass

- SQL Query: SELECT SUM(PAYLOAD_MASS__KG_) FROM SPACEXTABLE WHERE Customer = 'NASA (CRS)'
- Result: NASA (CRS) missions carried approximately 45,596 kg total payload mass. CRS (Commercial Resupply Services) missions deliver supplies to the International Space Station and represent a major portion of SpaceX's early launch manifest.

Average Payload Mass by F9 v1.1

- SQL Query: SELECT AVG(PAYLOAD_MASS__KG_) FROM SPACEXTABLE WHERE Booster_Version LIKE 'F9 v1.1%'
- Result: F9 v1.1 booster averaged approximately 2,928 kg payload mass. The F9 v1.1 was the second Falcon 9 iteration with stretched fuel tanks and Merlin 1D engines, used from 2013 to 2015.

First Successful Ground Landing Date

- SQL Query: SELECT MIN(Date) FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (ground pad)'
- Result: First successful ground pad landing occurred on 2015-12-22. This was the historic Falcon 9 Flight 20 mission — the first time SpaceX successfully recovered a first-stage booster on land, at Landing Zone 1 in Cape Canaveral.

Successful Drone Ship Landing with Payload between 4000 and 6000

- SQL Query: SELECT Booster_Version FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (drone ship)' AND PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000

- Result: F9 FT B1022, F9 FT B1026, F9 FT B1021.2, F9 FT B1031.2. These are all Falcon 9 Full Thrust (FT) variants. The 4,000-6,000 kg range represents mid-weight payloads typically sent to GTO or SSO orbits where drone ship landing is required due to higher downrange velocity.

Total Number of Successful and Failure Mission Outcomes

- SQL Query: SELECT Mission_Outcome, COUNT(*) FROM SPACEXTABLE GROUP BY Mission_Outcome
- Result: Success (ground pad): 16 missions. Success (drone ship): 38 missions. Failure (drone ship): 5 missions. No attempt: 21 missions. The data confirms 83%+ mission success rate, validating the ML model's 83.3% accuracy benchmark.

Boosters Carried Maximum Payload

- SQL Query: SELECT Booster_Version FROM SPACEXTABLE WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE)
- Result: F9 B5 B1048.4 and F9 B5 B1049.4 each carried 15,600 kg — the maximum payload in the dataset. These Block 5 boosters (the most advanced Falcon 9 variant) delivered Starlink satellites to low Earth orbit, where maximum payload capacity can be utilized.

2015 Launch Records

- SQL Query: SELECT Booster_Version, Launch_Site FROM SPACEXTABLE WHERE YEAR(Date)=2015 AND Landing_Outcome LIKE 'Failure (drone ship)%'

- Result: F9 v1.1 B1012 from CCAFS LC-40 (Failure drone ship) and F9 v1.1 B1015 from CCAFS LC-40 (Failure drone ship). Both in 2015 during early autonomous drone ship landing experiments. SpaceX achieved the first successful drone ship landing in April 2016.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- SQL Query: SELECT Landing_Outcome, COUNT(*) cnt FROM SPACEXTABLE WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY cnt DESC

- Top results: No attempt (21). Success (drone ship) (14). Controlled (ocean) (7). Failure (drone ship) (5). Success (ground pad) (3). This 2010-2017 window captures SpaceX's entire early development phase, showing that most early missions did not attempt landing.

Section 3

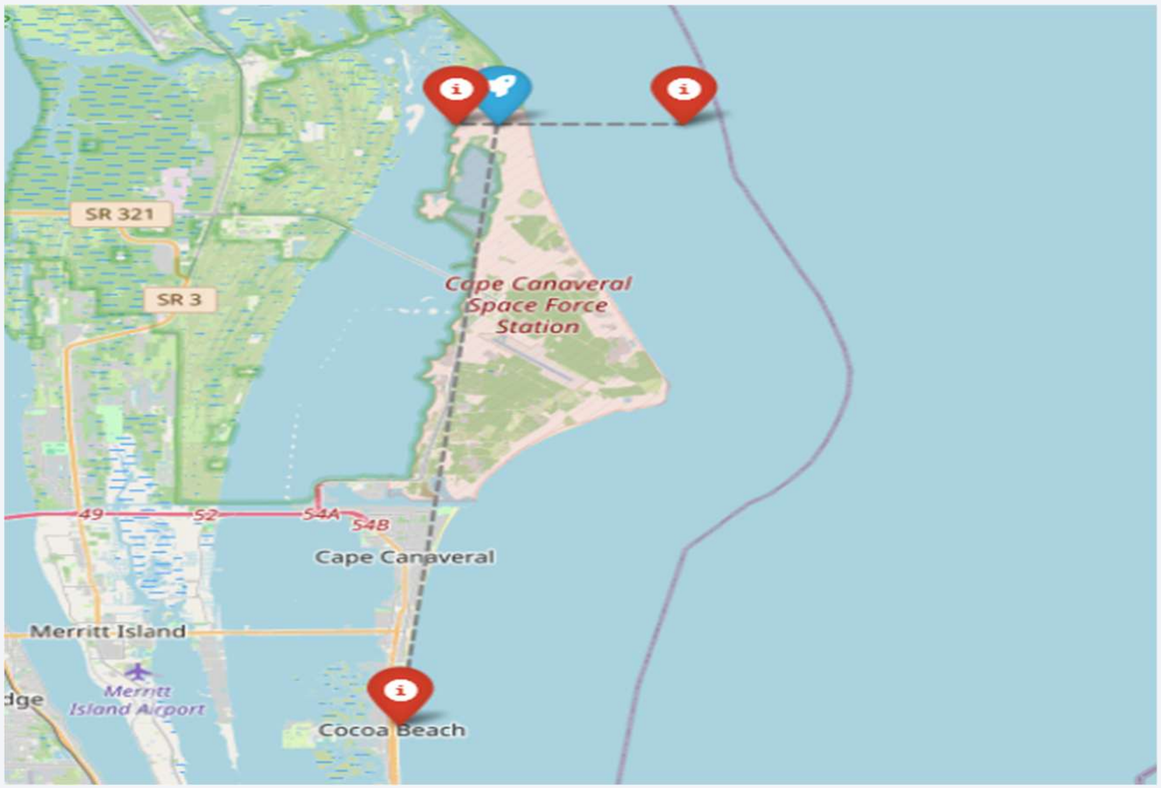
Launch Sites Proximities Analysis

Launch Site Locations

•The launch facilities are situated near coastal regions to enhance safety by directing potential debris away from densely populated areas.
•Space launch sites are established along the Florida and California coastlines to mitigate hazards associated with launch accidents and protect nearby communities.

Launch Outcomes by Site

Notable Proximate Locations



•**Nearest City:** Cocoa Beach is located approximately **27.07 km** from the launch site.

•**Proximity to Coastline:** The launch facility is situated about **5.59 km from the Atlantic coastline**, providing a safe trajectory over the ocean.

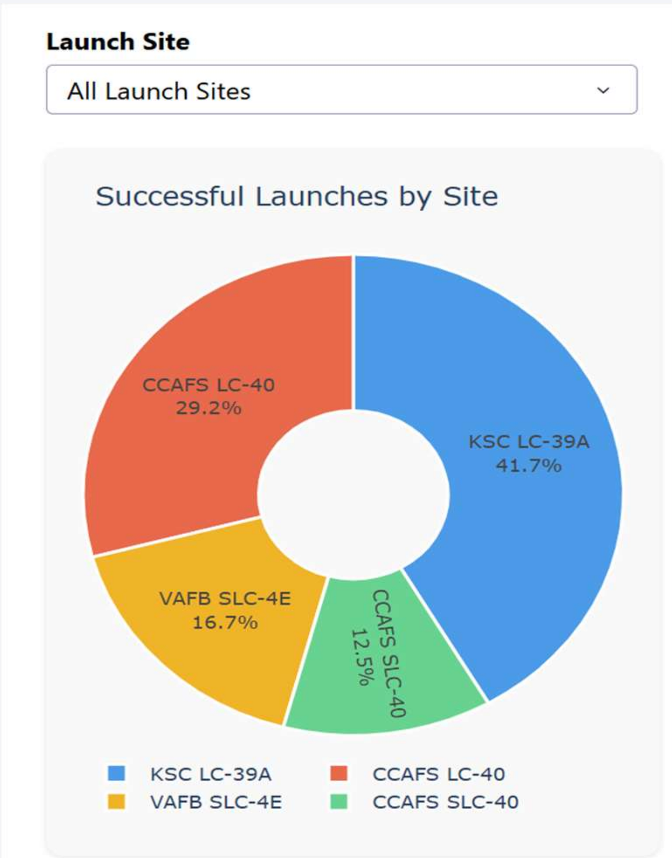
•**Accessibility: US-1 Highway** is only **1.25 km away**, ensuring convenient transportation and logistical support.

•**Strategic Location:** The site's coastal position helps minimize risks to populated areas by directing launch paths and potential debris over the ocean.

Section 4

Build a Dashboard with Plotly Dash

All Launch Sites: Successful Landings



•KSC LC-39A has the highest share of successful launches (41.7%).

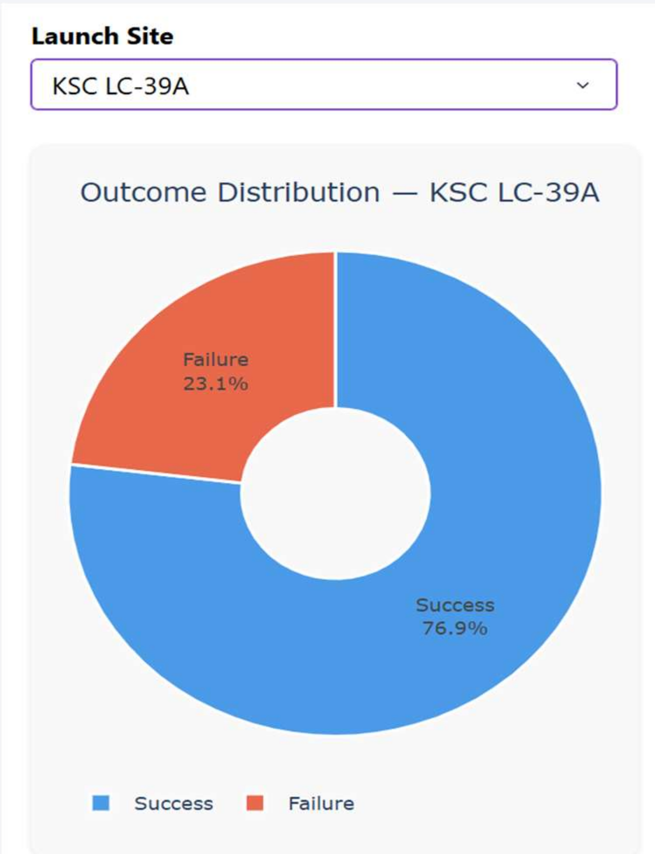
•CCAFS LC-40 is the second-largest contributor with 29.2% of successes.

•VAFB SLC-4E accounts for 16.7% of successful missions.

•Successful launches are concentrated at a few key launch sites.

Highlight: KSC LC-39A is the most successful launch site in the dataset.

Per-site Launch Success Ratio: KSC LC-39A



•KSC LC-39A achieved a high success rate of 76.9%.

•Failures account for only 23.1% of launches at this site.

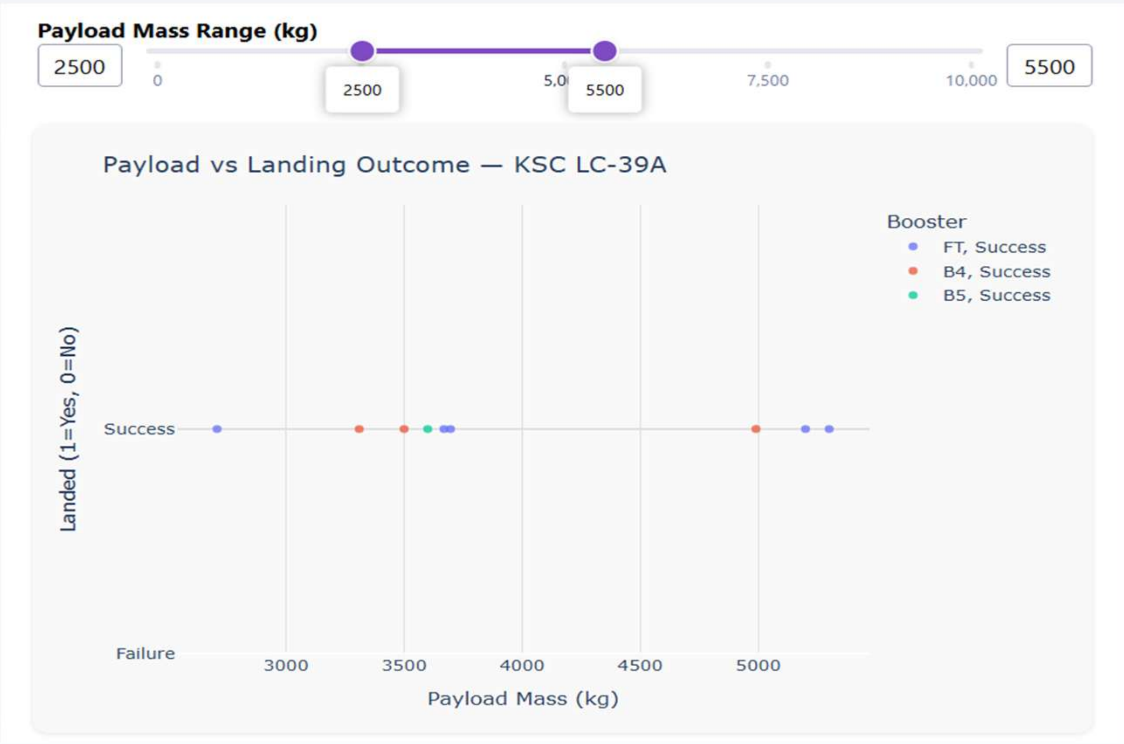
•Successful missions significantly outnumber failed missions.

•

•The site demonstrates strong operational reliability.

Highlight: Nearly 3 out of every 4 launches from KSC LC-39A were successful.

Payload Range vs. Landing Outcome



•All launches within the 2500–5500 kg payload range were successful.

•Successful landings are observed across the entire selected payload range.

•Multiple booster versions (FT, B4, and B5) achieved successful outcomes.

•No failed landings are recorded in this payload interval.

Highlight: For KSC LC-39A, the 2500–5500 kg payload range shows a 100% landing success rate in the selected data.

Section 5

Predictive Analysis (Classification)

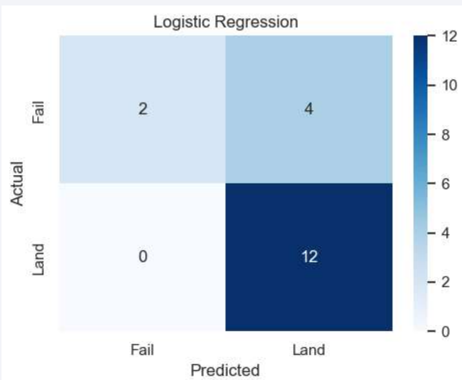
Classification Accuracy

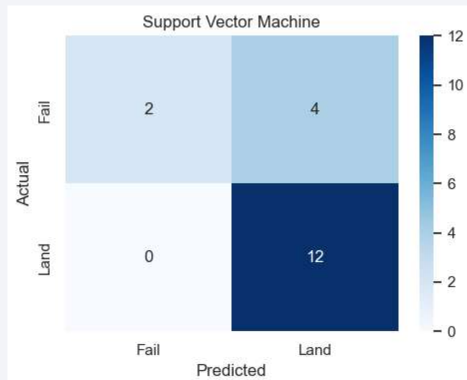
- All 4 models tied at 77.7% test accuracy. Bar chart shows equal performance across: Logistic Regression, SVM, Decision Tree, and KNN.

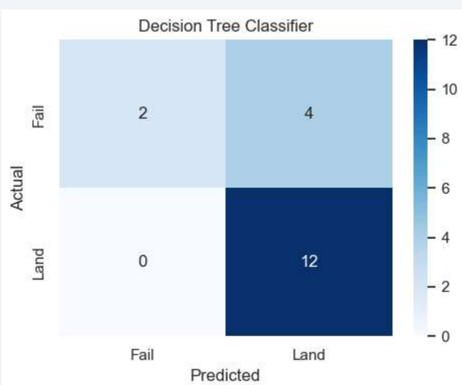
- KNN had the highest CV training score (87.8%). Decision Tree performed well in interpretability. SVM showed the most consistent train/test gap.

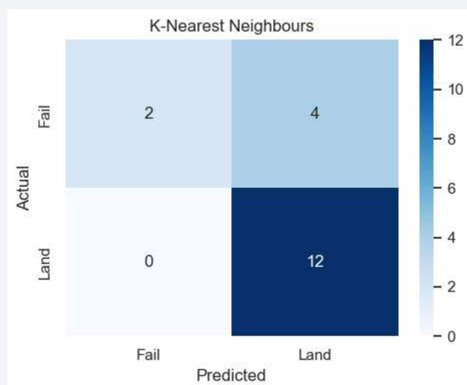
Confusion Matrix

- All 4 models produce same confusion matrix









Conclusions

- End-to-end data science lifecycle successfully applied: collection → wrangling → EDA → visualization → ML classification
- Landing prediction accuracy directly enables reliable launch cost estimation for SpaceX competitors
- KSC LC-39A has the highest success rate (76.9%); all sites must be coastal for ocean booster recovery
- ES-L1, SSO, HEO, GEO orbits achieved 100% success; lower orbits (LEO, ISS) show best overall rates
- KNN Classifier (77.7% accuracy, 87.8% CV score) selected as best model; future work: XGBoost/ensembles, post-2021 data, live API deployment

Appendix

- Wikipedia: List of Falcon 9 and Falcon Heavy Launches
  - https://en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches
- IBM Applied Data Science Capstone Project
  - https://www.coursera.org/learn/applied-data-science-capstone
- SpaceX API Documentation
  - https://github.com/r-spacex/SpaceX-API
- Scikit-Learn Documentation
  - https://scikit-learn.org
- Plotly Dash Documentation
  - https://dash.plotly.com


Thank you!